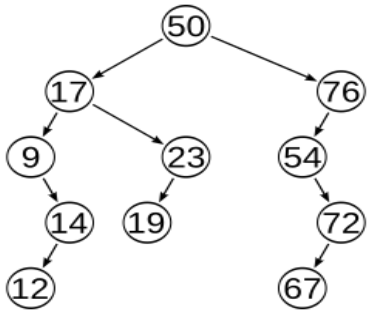
Tutorial Quiz
Set A

Instruction: **Max. Marks: 15**
Write in clear handwriting.

Q1.The following unbalanced **AVL Tree** has the keys (in insertion order):



i) **Balance the tree** by performing the necessary rotations and show the AVL tree after each rotation step. **[1 mark]**
ii) **Insert** the following values in the given order: {55,35, 85}. After each insertion, draw the new AVL tree. Let 'm' be total number rotations used for inserting above 3 values what is the value of 'm'? **[2 marks]**
iii) **Delete** the following values from the tree: {50, 17}. After each deletion, draw the updated AVL tree. **[2 marks]**

Q2. The following keys represent the elements of a MAX-HEAP stored in array form:
{90, 75, 60, 68, 55, 50, 42, 30}

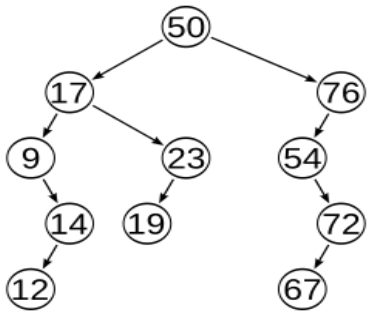
1. Insert the values 65 and 80 (in this order) into the heap and draw the new heap tree. **[2 marks]**
2. Delete two elements (one by one, starting from the root each time) from the resulting heap from the previous part and draw the final heap tree. **[2 marks]**
3. For the final tree answer the following questions: **[1 mark]**
    i. Height of the tree.
    ii. Depth of the tree.

Q3. The following unsorted array is given (array indices start at 1 for heap explanations):
{47, 3, 89, 21, 56, 73}
Sort this array using **heapsort (max-heap)**. Do every step and explain what you are doing at each step. **[5 marks]**

Tutorial Quiz
Set B

Instructions: **Max. Marks: 15**
Write in clear handwriting.

Q1.The following unbalanced **AVL Tree** has the keys (in insertion order):



i) **Balance the tree** by performing the necessary rotations and show the AVL tree after each rotation step. **[1 mark]**
ii) **Insert** the following values in the given order: {46, 6, 66}. After each insertion, draw the new AVL tree. Let 'm' be total number rotations used for inserting above 3 values what is the value of 'm'? **[2 marks]**
iii) **Delete** the following values from the tree: {76, 9}. After each deletion, draw the updated AVL tree. **[2 marks]**

Q2. The following keys represent the elements of a MAX-HEAP stored in array form:
{70, 58, 63, 42, 37, 49, 55, 28}

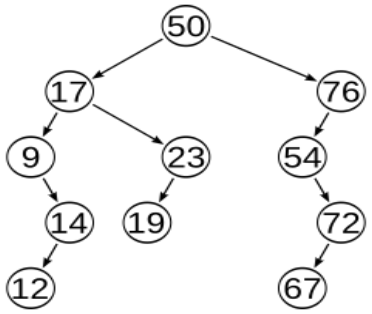
1. Insert the values 64 and 75 (in this order) into the heap and draw the new heap tree. **[2 marks]**
2. Delete two elements (one by one, starting from the root each time) from the resulting heap from the previous part and draw the final heap tree. **[2 marks]**
3. For the final tree answer the following questions: **[1 mark]**
    i. Height of the tree.
    ii. Depth of the tree.

Q3. The following unsorted array is given (array indices start at 1 for heap explanations):
{102, 15, 87, 43, 68, 99, 2, 76, 54, 31}
Sort this array using **heapsort (max-heap)**. Do every step and explain what you are doing at each step. **[5 marks]**

Tutorial Quiz
Set C

Instructions: **Max. Marks: 15**
Write in clear handwriting.

Q1.The following unbalanced **AVL Tree** has the keys (in insertion order):



i) **Balance the tree** by performing the necessary rotations and show the AVL tree after each rotation step. **[1 mark]**
ii) **Insert** the following values in the given order: {13, 78, 20}. After each insertion, draw the new AVL tree. Let 'm' be total number rotations used for inserting above 3 values what is the value of 'm'? **[2 marks]**
iii) **Delete** the following values from the tree: {54, 17}. After each deletion, draw the updated AVL tree. **[2 marks]**

Q2. The following keys represent the elements of a MAX-HEAP stored in array form:
{95, 85, 80, 70, 60, 55, 50, 40}

1. Insert the values 77 and 88 (in this order) into the heap and draw the new heap tree. **[2 marks]**
2. Delete two elements (one by one, starting from the root each time) from the resulting heap from the previous part and draw the final heap tree. **[2 marks]**
3. For the final tree answer the following questions: **[1 mark]**
    i. Height of the tree.
    ii. Depth of the tree.

Q3. The following unsorted array is given (array indices start at 1 for heap explanations):
{250, 11, 199, 75, 180, 5, 123, 67, 214, 88}
Sort this array using **heapsort (max-heap)**. Do every step and explain what you are doing at each step. **[5 marks]**